

The macroeconomic effects of Stablecoins

Lorenzo Forni

September 2025

1 Money multiplier derivation

We start from the stylized balance sheets.

Central Bank (CB).

- Assets: government bonds, loans to banks L_{CB} .
- Liabilities: banknotes (currency held by the public) C , reserves (banks' deposits at the CB) R .

Commercial Banks.

- Assets: loans to the private sector and to the government, financial assets, reserves at the CB R .
- Liabilities: deposits of the public D , borrowing from CB L_{CB} , bonds issued, equity.

Two key aggregates:

$$M \equiv C + D \quad (\text{money supply: currency} + \text{deposits}),$$

$$MB \equiv C + R \quad (\text{monetary base: currency} + \text{reserves}).$$

Behavioral Ratios

We define:

$$c \equiv \frac{C}{D} \quad (\text{currency/deposit}), \quad r \equiv \frac{R}{D} \quad (\text{reserve/deposit}).$$

Then

$$C = cD, \quad R = rD.$$

Derivation

$$M = C + D = cD + D = (1 + c)D,$$

$$MB = C + R = cD + rD = (c + r)D.$$

The money multiplier m is defined as:

$$m \equiv \frac{M}{MB}.$$

Substituting:

$$m = \frac{(1 + c)D}{(c + r)D} = \frac{1 + c}{c + r}.$$

$$m = \frac{1 + c}{c + r}$$

Extensions

If we split the reserve ratio into required and excess reserves, $r = rr + er$, then:

$$m = \frac{1 + c}{c + rr + er}.$$

Interpretation

- Higher currency preference c reduces m .
- Higher reserve ratio r (whether due to regulation or excess reserves) also reduces m .
- The asset mix of the CB (government bonds vs. loans) or non-deposit funding of banks affect the level of MB , but not the formula for m .

2 Extension: Stablecoin issuer and the short term effects on the money multiplier

2.1 Stablecoins and the level of money

Benchmark (no stablecoins). With deposits D_0 , currency C_0 , and reserves $R_0 = rD_0$, we have

$$M_0 = C_0 + D_0, \quad MB_0 = C_0 + R_0 = C_0 + rD_0.$$

With stablecoins. Suppose the public acquires S of stablecoins by reducing deposits. Currency is unchanged, while deposits and reserves fall proportionally:

$$D_1 = D_0 - S, \quad C_1 = C_0, \quad R_1 = rD_1 = r(D_0 - S).$$

Conventional money becomes

$$M_1 = C_1 + D_1 = M_0 - S, \quad \Delta M = -S.$$

If stablecoins are included as money-like instruments,

$$M_1^* = C_1 + D_1 + S = M_0, \quad \Delta M^* = 0.$$

More generally, if stablecoins carry liquidity weight $\phi \in [0, 1]$,

$$M_1^\phi = M_0 - (1 - \phi)S, \quad \Delta M^\phi = -(1 - \phi)S.$$

2.2 Stablecoins and the Money Multiplier

Benchmark.

$$m_0 = \frac{M_0}{MB_0} = \frac{C_0 + D_0}{C_0 + rD_0}.$$

After stablecoins. On impact,

$$M_1 = C_0 + (D_0 - S), \quad MB_1 = C_0 + r(D_0 - S).$$

Thus

$$m_1 = \frac{M_1}{MB_1} = \frac{C_0 + D_0 - S}{C_0 + rD_0 - rS}$$

Comparative statics.

- Both money M and the monetary base MB shrink when deposits fall.
- The effect on the multiplier m depends on r :

$$m_1 - m_0 = \frac{C_0 + D_0 - S}{C_0 + rD_0 - rS} - \frac{C_0 + D_0}{C_0 + rD_0}.$$

- For small S , the approximate change is

$$\Delta m \approx \frac{(r - 1)S}{(C_0 + rD_0)^2} C_0.$$

Since $r < 1$, the multiplier typically *falls*, but less sharply than in the case where reserves are fixed.

Extended aggregates. If we include stablecoins,

$$m_1^* = \frac{M_1^*}{MB_1} = \frac{M_0}{C_0 + rD_0 - rS}$$

so m^* rises, because the numerator is constant while the monetary base is smaller.

Key implication. When reserves adjust proportionally with deposits ($R = rD$), the stablecoin substitution has two effects:

1. Conventional money M falls one-for-one with S .
2. The monetary base MB also falls (since R shrinks with D).

The result is that the conventional multiplier falls, while the extended multiplier m^* rises.

3 Stablecoins fully invested in government bonds: impact on bond demand, bank lending, and reserves

Baseline (no stablecoins). Let

D_0 : Deposits held by the public at commercial banks,

$L_{P,0}$: Commercial banks' loans to the private sector,

G_0^{dem} : Total demand for government bonds (private sector + banks, excl. CB),

C_0 : Currency in circulation,

$R_0 = rD_0$: Bank reserves (proportional to deposits, with $r \in (0, 1)$).

Introduce stablecoins. A stablecoin issuer sells S units of stablecoins to the public, funded by deposits withdrawals, and invests the proceeds fully in government bonds. Bond purchases are in the secondary market and on impact don't change the overall offer of government bonds B . Because $R = rD$, a fall in deposits mechanically reduces reserves:

$$D_1 = D_0 - S, \quad R_1 = rD_1 = r(D_0 - S) = R_0 - rS.$$

Government bond demand. Stablecoin collateral purchases raise private demand for government bonds by S :

$$G_1^{\text{dem}} = G_0^{\text{dem}} + S \quad \Rightarrow \quad \boxed{\Delta G^{\text{dem}} = S}.$$

Commercial-bank loans to the private sector. Assume the deposit outflow is accommodated entirely by reducing loans to the private sector:

$$L_{P,1} = L_{P,0} - S \quad \Rightarrow \quad \boxed{\Delta L_P = -S}.$$

Reserves and monetary base. With $R = rD$,

$$\boxed{\Delta R = R_1 - R_0 = -rS}.$$

If the monetary base is $MB = C + R$ and currency is unchanged on impact, then

$$\boxed{\Delta MB = \Delta C + \Delta R = -rS}.$$

Balance-sheet consistency (sketch).

- *Public:* swaps S of bank deposits for S of stablecoins (asset composition shift).
- *Stablecoin issuer:* issues a liability S (stablecoins) and acquires S of government bonds (asset).
- *Commercial banks (consolidated):* deposits fall by S (liability); reserves fall by rS (asset) due to $R = rD$; the remaining asset-side adjustment of $S - rS$ is made by reducing private loans L_P one-for-one with the deposit outflow:

$$\Delta \text{Assets} = \underbrace{\Delta R}_{-rS} + \underbrace{\Delta L_P}_{-S} = -S - rS, \quad \Delta \text{Liabilities} = \underbrace{\Delta D}_{-S},$$

and the difference is absorbed by equity/other items as usual in partial equilibrium. Under our maintained assumption, we record the lending effect as $\Delta L_P = -S$.

Remarks.

- The contraction in reserves is proportional to the fall in deposits, with $\Delta R = -rS$.
- Because part of the adjustment occurs through reserves, the reduction in commercial-bank lending to the private sector could be smaller than the full deposit outflow. Specifically,

$$\Delta L_P = -(1 - r)S,$$

so only the share of the deposit contraction not absorbed by reserves might translate into lower private lending.

- The overall effect of stablecoin issuance might therefore be a reallocation of funds: an increase of S in demand for government bonds, a reduction of $(1 - r)S$ in private-sector loans, and a fall of rS in reserves.
- Let's maintain this assumption in the remainder of this note.

4 Macro implications of stablecoins: a simple two-market model

Environment and shock

Baseline (no stablecoins): deposits D_0 , currency C_0 , reserves $R_0 = rD_0$, bank loans to the private sector $L_{P,0}$, and outstanding government bonds B .

Introduce stablecoins S purchased by the public out of deposits. The issuer invests the proceeds fully in government bonds. On impact:

$$D_1 = D_0 - S, \quad R_1 = rD_1 = r(D_0 - S), \quad \Delta L_P = -(1-r)S.$$

Thus both deposits and reserves fall with S , while bank lending to the private sector contracts by the remaining share $(1-r)S$.

Bond and loan market blocks

Government bonds. Stablecoin demand shifts the private demand schedule for government bonds up by S :

$$Q_g^d = a_g + b_g r_g + S, \quad b_g > 0.$$

Market clearing $B = Q_g^d$ gives

$$r_g = \frac{B - a_g - S}{b_g}, \quad \boxed{\Delta r_g = -\frac{1}{b_g}S < 0.}$$

Private loans. Loan demand is $L_P^d = a_p - b_p r_p$ with $b_p > 0$. Bank loan supply falls from $\bar{L}$ to $\bar{L} - (1-r)S$ because part of the deposit outflow is absorbed by reserves. Market clearing:

$$a_p - b_p r_p = \bar{L} - (1-r)S \implies r_p = \frac{a_p - \bar{L} + (1-r)S}{b_p}, \quad \boxed{\Delta r_p = \frac{1-r}{b_p}S > 0.}$$

Real side and fiscal block

Investment and output. With $I = I_0 - \eta r_p$ ($\eta > 0$),

$$\Delta I = -\eta \Delta r_p = -\frac{\eta}{b_p}(1-r)S.$$

If output responds to investment with multiplier $\kappa > 0$,

$$\Delta Y = \kappa \Delta I = -\underbrace{\left(\frac{\kappa \eta}{b_p}\right)}_{\psi}(1-r)S, \quad \boxed{\Delta Y = -\psi(1-r)S, \quad \psi = \frac{\kappa \eta}{b_p} > 0.}$$

Taxes and the fiscal deficit. With proportional taxes $T = \tau Y$,

$$\Delta T = \tau \Delta Y = -\tau \psi(1-r)S.$$

Government interest outlays are $r_g B$, so

$$\Delta \text{Int}_g = B \Delta r_g = -\frac{B}{b_g} S.$$

Hence, with fixed primary spending G ,

$$\Delta \text{Def} = \Delta \text{Int}_g - \Delta T = \left(\tau \psi(1-r) - \frac{B}{b_g} \right) S.$$

$$\Delta \text{Def} = \left(\tau \psi(1-r) - \frac{B}{b_g} \right) S$$

Interpretation and threshold condition

- **Yields:** Stablecoins reduce the government bond yield ($\Delta r_g < 0$) and raise the private loan rate ($\Delta r_p > 0$).
- **Real activity:** Higher r_p lowers investment and output: $\Delta I = -(\eta/b_p)(1-r)S$, $\Delta Y = -\psi(1-r)S$.
- **Fiscal balance:** Two channels:
 1. Lower r_g reduces debt service: $-\frac{B}{b_g} S$.
 2. Lower output reduces tax revenues: $+\tau \psi(1-r)S$.

The deficit rises if the tax effect dominates:

$$\tau \psi(1-r) > \frac{B}{b_g} \implies \Delta \text{Def} > 0.$$

Notes

- Because $R = rD$, reserves fall automatically when deposits fall, absorbing part rS of the adjustment. The remaining $(1-r)S$ is borne by private lending.
- If banks can offset deposit outflows with wholesale funding or asset sales, the contraction in L_P is smaller than $(1-r)S$, muting Δr_p and the output/tax channel.